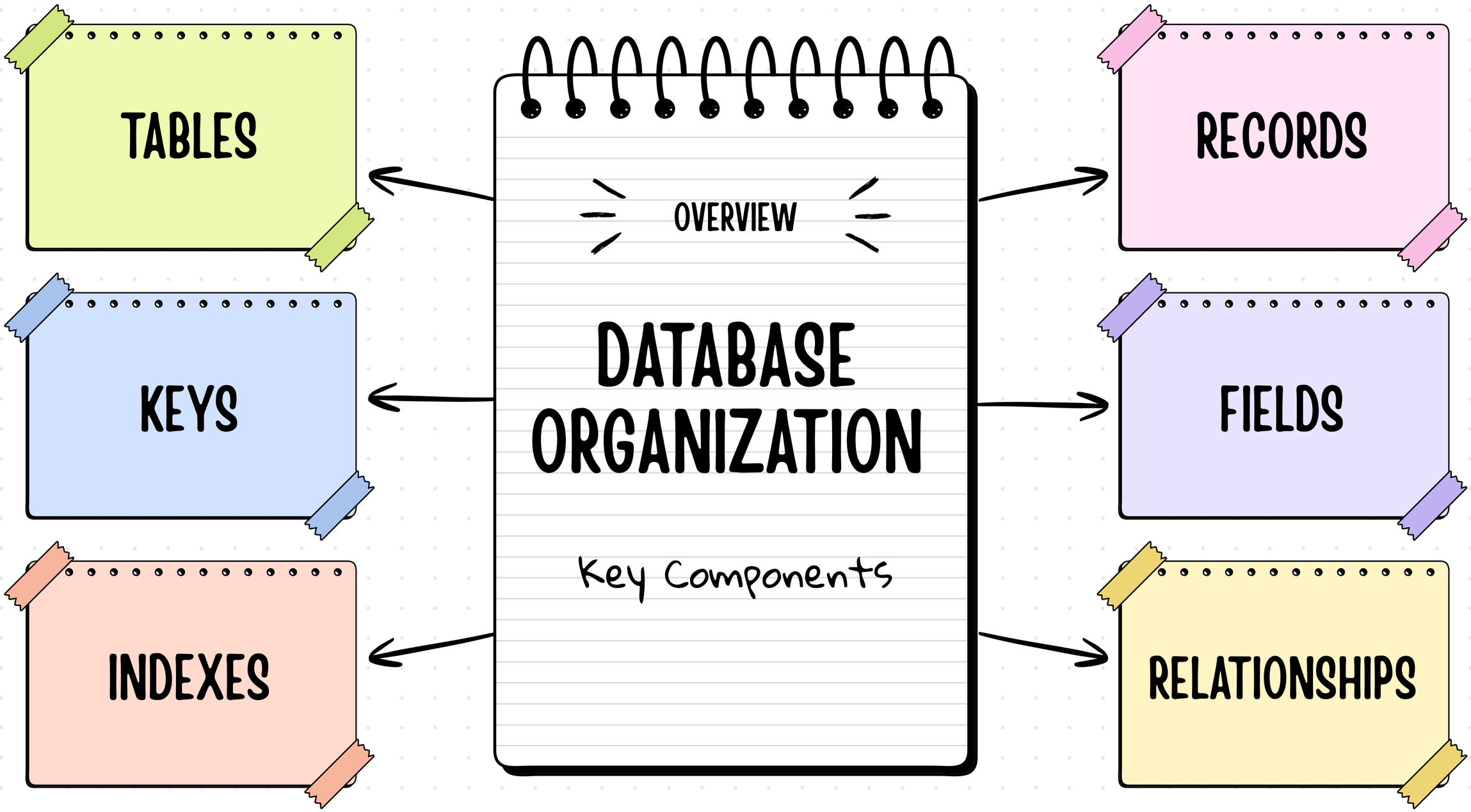

TABLES
KEYS
INDEXES
OVERVIEW
DATABASE ORGANIZATION
Key Components
RECORDS
FIELDS
RELATIONSHIPS

DATA ORGANIZATION FLOW

Raw Data

Unstructured information enters the database system. This includes text, numbers, dates, and other data types collected from various sources before processing.

01

Tables

Data is organized into tables - structured grids with rows and columns. Each table represents a specific entity like customers, products, or orders.

02

Records

Each row in a table is a record representing one complete entry. A customer record contains all information about a single customer.

03

Fields

Fields are the columns that define what data each record contains. Examples include name, email, date, and ID fields with specific data types.

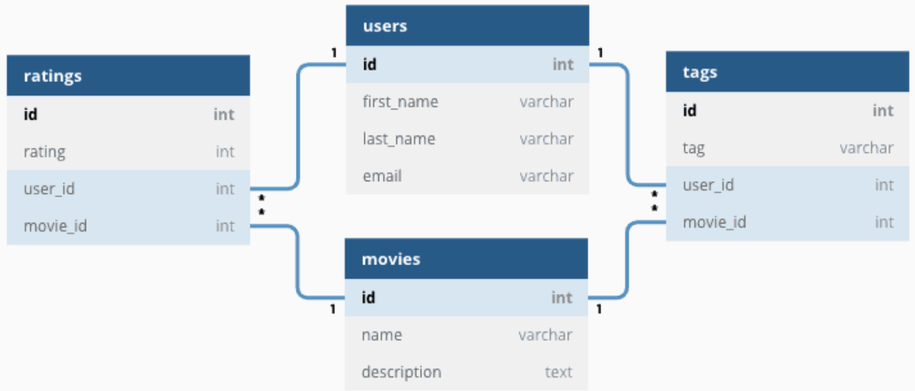
04

Indexed DB

Indexes are created on frequently searched fields for faster retrieval. The organized database enables efficient queries and data management.

05

e_id	e_name	e_salary	e_age	e_gender	e_dept
1	Sam	95000	45	Male	Operations
2	Bob	80000	21	Male	Support
3	Anne	125000	25	Female	Analytics
4	Julia	73000	30	Female	Analytics
5	Matt	159000	33	Male	Sales
6	Jeff	112000	27	Male	Operations




```
select * from sys.allocation_units
```

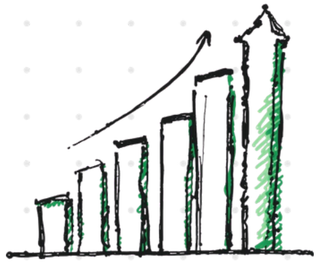
118 %

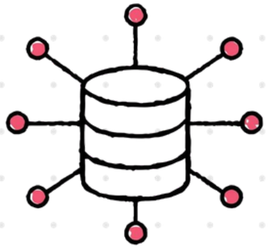
Results | Messages

	allocation_unit_id	type	type_desc	container_id	data_space_id	total_pages	used_pages	data_pages
82	844424932884480	1	IN_ROW_DATA	844424932884480	1	0	0	0
83	844424933408768	1	IN_ROW_DATA	844424933408768	1	2	2	1
84	844424935768064	1	IN_ROW_DATA	844424935768064	1	0	0	0
85	1125899909070...	1	IN_ROW_DATA	1125899909070...	1	17	10	8
86	7177611906514...	2	LOB_DATA	281474980642816	1	12	6	0
87	7205759403799...	1	IN_ROW_DATA	281474980511744	1	2	2	1
88	7205759403805...	1	IN_ROW_DATA	562949957222400	1	2	2	1
89	7205759403819...	1	IN_ROW_DATA	281474983067648	1	2	2	1
90	7205759403825...	1	IN_ROW_DATA	562949959778304	1	2	2	1

customer_id	first_name	last_name	phone	email	street	city	state	zip_code
1	Debra	Burks	NULL	debra.burks@yahoo.com	9273 Thorne Ave.	Orchard Park	NY	14127
2	Kasha	Todd	NULL	kasha.todd@yahoo.com	910 Vine Street	Campbell	CA	95008
3	Tameka	Fisher	NULL	tameka.fisher@aol.com	769C Honey Creek St.	Redondo Beach	CA	90278
4	Daryl	Spence	NULL	daryl.spence@aol.com	988 Pearl Lane	Uniondale	NY	11553
5	Charolette	Rice	(916) 381-6003	charolette.rice@msn.com	107 River Dr.	Sacramento	CA	95820
6	Lyndsey	Bean	NULL	lyndsey.bean@hotmail.com	769 West Road	Fairport	NY	14450
7	Latasha	Hays	(716) 986-3359	latasha.hays@hotmail.com	7014 Manor Station Rd.	Buffalo	NY	14215
8	Jacquline	Duncan	NULL	jacquline.duncan@yahoo.com	15 Brown St.	Jackson Heights	NY	11372
9	Genoveva	Baldwin	NULL	genoveva.baldwin@msn.com	8550 Spruce Drive	Port Washington	NY	11050
10	Pamelia	Newman	NULL	pamelia.newman@gmail.com	476 Chestnut Ave.	Monroe	NY	10950

BENEFITS OF DATABASE ORGANIZATION





Fast Queries

Well-organized databases with proper indexing enable lightning-fast data retrieval. Queries execute efficiently, reducing wait times and improving application performance significantly.

Data Integrity

Structured organization ensures data accuracy and consistency. Constraints, relationships, and validation rules maintain reliable information across all database operations.

Scalability

Properly organized databases grow seamlessly with your needs. Good structure allows for easy expansion, handling increased data volumes without performance degradation.

Easy Maintenance

Clear organization simplifies updates, backups, and troubleshooting. Database administrators can quickly locate issues and implement changes with minimal disruption.

Reduced Redundancy

Normalization eliminates duplicate data storage. This saves space, prevents inconsistencies, and ensures that updates only need to be made in one location.